

Andreina Cota Vidaurre



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<https://github.com/andreina-covi>

Education

- 2022 - present PhD Candidate
UC - Pontificia Universidad Católica de Chile
DCC - Departamento Ciencia de la Computación
- 2018 Informatic Engineer
UMSS - Universidad Mayor de San Simón (5 years program)
Ranking: First of the class

Professor

- 2024 - 2025 Modelos multimodales
CENIA and Universidad de Tarapacá UTA
Diplomado en IA generativa y modelos de lenguaje

Delivered comprehensive instruction on multimodal models, with a specific focus on the BLIP architecture. The module encompassed the model's architectural design, core contributions, and practical implementation, facilitated through hands-on notebook demonstrations. Furthermore, I provided a critical analysis of BLIP's strengths, limitations, and ongoing research challenges, enriching student understanding with dynamic, real-world inference examples.

Assistant Professor

- 2025 Capstone Project Tutor (AFG)
UC - Pontificia Universidad Católica de Chile
Magíster online en Ciencia de Datos

Provided academic supervision for the final capstone (AFG), supporting teams in project planning and execution, monitoring progress on a consistent cadence, and ensuring deliverables met program expectations.

2025 - present	Course Assistant UC - Pontificia Universidad Católica de Chile Aprendizaje Profundo, Magíster en Ciencia de Datos - Coursera Provided comprehensive student guidance through meticulous review of assignments and projects, delivering constructive feedback to enhance their academic growth.
2024	Course tutor UC - Pontificia Universidad Católica de Chile DCC - Python para Machine Learning - Clase Ejecutiva UC Facilitated student success through comprehensive online instruction, rapid query resolution, and rigorous project evaluations, delivering targeted feedback to optimize practical outcomes.
2023 - present	Correction Teaching Assistant UC - Pontificia Universidad Católica de Chile Diplomado en Inteligencia Artificial - Educación Profesional UC Enhanced student learning through meticulous review of practical assignments, offering constructive guidance to foster academic development.
2023 - present	Course Assistant UC - Pontificia Universidad Católica de Chile DCC - Ingeniería UC Guided student project development through detailed assignment reviews and feasibility assessments, providing key strategic guidance to ensure successful project execution.

--- Research Project Participation

Title	Drone-based Early Wildfire Detection
Description	Project under the direction Ph.D. Eduardo Di Santi. I was part of the "Software Factory" team, which was responsible to develop a drone-based system for detecting wildfires. In particular, I was responsible for developing a fire detection system through HOG (Histogram of Oriented Gradients) and SVM (Support Vector Machine).

Experience

- 2022 CENIA(Centro Nacional de Inteligencia Artificial)
Machine Learning Engineer
My main responsibilities include:
- Design and developing Machine Learning systems
 - Data analysis
- 2020-2022 ObjectProfile SpA.
Software development and Research
Supervisor: Alexandre Bergel, PhD
My main responsibilities include:
- Apply Machine Learning to infer type variables from dynamically typed language source code
 - Apply Genetic Programming to automatically create method argument validators
- 2018-2020 Semantics S.R.L.
Software development and Research
Supervisor: Juan Pablo Sandoval, PhD
My main responsibilities include:
- Build an interpreter that resolves GraphQL queries in the Pharo programming Language
 - Use Genetic Algorithms to automatically generate unit tests
 - Propose a visualization tool to analyse the population evolution of Genetic Algorithms
- 2017 - 2018 PIL Andina S.A.
Software developer
My main responsibilities include:
- Develop and maintain a sales and distribution system, using Java and Spring
 - Develop and maintain the sales and distribution system of mobile app, using Android
 - Support for system users
 - Prepare of reports based on database for different areas of the company

Publications

- 2020 TestEvoViz: Visual Introspection for Genetically-Based Test Coverage Evolution
Andreina Cota Vidaure, Evelyn Cusi Lopez, Juan Pablo Sandoval Alcocer, Alexandre Bergel.
Proceedings of 8th IEEE Working Conference on Software Visualization (VIS-SOFT '20).
- 2018 Deviation Testing: A Test Case Generation Technique for GraphQL APIs
Daniela Meneses Vargas, Alison Fernandez Blanco, Andreina Cota Vidaurre, Juan Pablo Sandoval Alcocer, Milton Mamani Torres, Alexandre Bergel, Stéphane Ducasse.
Proceedings of International Workshop on Smalltalk Technologies (IWST '18).
- 2016 Visual automatic fire detection through the Histogram of Oriented Gradients and Support Vector Machine
Andreina Cota, Evelyn Cusi, Carlos Escobar, Kevin Orellana, Patricia Rogriguez.
UMSA: Revista Investigación y Tecnología

Skills

- ✦ Machine Learning, Deep Learning, Natural Language Processing, Computer Vision, Evolutionary Algorithms, Multimodal models
- ✦ Relational (MySQL, SQLite) and NoSQL (MongoDB) Databases
- ✦ Python, Java, Pharo
- ✦ Pharo IDE, Roassal, VSCode
- ✦ Pandas, Numpy, SciPy, Scikit-learn, Matplotlib, OpenCV, Jupyter, Django
- ✦ Linux (Ubuntu, Debian), Windows, MacOS
- ✦ Git, Octave, Latex, Docker

Additional Courses

- 2021 **Machine Learning**
AI Saturdays La Paz
- 2020 **Natural Language Processing**
Specialized Coursera program about NLP that includes four courses
- 2019 **Introduction to Neural Networks, Genetic Algorithms, and NeuroEvolution**

	Tutorial taken in Bolivian Summer School event, UCB (Universidad Catolica Boliviana), given by Ph.D. Alexandre Bergel
2019	Reinforcement Learning for Video Games
	Bolivian Summer School, UCB given by Ph.D. Julio Godoy
2019	Deep Learning
	A Specialized Coursera program that includes five courses
2016	Object Detection
	Coursera's course from Universitat Autònoma de Barcelona

Other

2014	ICPC Bolivia
	Participation in the International Collegiate Programming Contest - South America/South Regional
2025	KHIPU.AI - Latin American AI Institute
	Participation in KHIPU, the most important artificial intelligence meeting in Latin America

References

Prof. Dr. Álvaro Soto
DCC, UC, Chile
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Prof. Dr. Alexandre Bergel
DCC, University of Chile, Chile
Email: abergel@dcc.uchile.cl
Website: <https://www.bergel.eu>

Prof. Dr. Juan Pablo Sandoval Alcocer
Pontificia Universidad Católica de Chile, Chile
Email: juanpablo.sandoval@ing.puc.cl
Website: <https://jpsandoval.github.io>

Languages

Spanish	Mothertongue
English	Conversational